



Year 12 Mathematics Extension 2
HSC ASSESSMENT TASK 1
Term 4 Week 6 2010

Name: _____

Teacher: _____

Friday 19th November

Set by: VUL

- Attempt **all** questions.
- All questions are of equal value.
- Marks may be deducted for insufficient, or illegible work.
- Only Board approved calculators (**excluding** graphic calculators) may be used
- Total possible mark is **40**
- **Begin each question on a new page.**
- **TIME ALLOWED : 60 minutes**

Question 1

Marks

- (a) Let $z = 3 - 4i$.
- (i) Find z^3 in the form $x + iy$ 1
- (ii) Find $z - 2\bar{z}$ in the form $x + iy$ 1
- (iii) Find $\frac{-i}{z}$ in the form $x + iy$ 2
- (b) Solve $z^2 = 5 + 12i$ for z giving your answers in the form $x + iy$ where x and y are real. 3
- (c) (i) Express $-1 + i$ in modulus-argument form. 1
- (ii) Hence evaluate $(-1 + i)^{12}$ 2
- (d) On separate Argand diagrams, sketch the locus of points z such that:
- (i) $\arg(z - i) = -\frac{\pi}{4}$ 2
- (ii) the inequalities $|z - 3 + i| \leq 5$ and $|z + 1| \leq |z - 1|$ both hold 3
- (iii) $|z| = \operatorname{Im}(z)$ 2
- (e) Prove that $12^n > 5^n + 7^n$ positive integers $n \geq 2$ 3
-

Question 2 **Start a new page.**

- (a) The points O, A, Z and C on the Argand diagram represent the complex numbers $0, 1, z$ and $z+1$ respectively, where $z = \cos \theta + i \sin \theta$ is any complex number of modulus 1, with $0 < \theta < \pi$.
- (i) Explain why $OACZ$ is a rhombus 1
- (ii) Show that $\frac{z-1}{z+1}$ is purely imaginary 2
- (iii) Find the modulus and argument of $z+1$ 2
- (b) (i) By considering $z^9 - 1$ as the difference of two cubes write $1 + z + z^2 + z^3 + z^4 + z^5 + z^6 + z^7 + z^8$ as a product of two polynomial factors with real coefficients, one of which is quadratic. 2
- (ii) Solve $z^9 - 1 = 0$ and determine the six solutions of $z^6 + z^3 + 1 = 0$ 2
- (iii) Hence show that $\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9}$ 2
- (c) Sketch the locus of the complex number $z = x + iy$ where $\arg \left[\frac{z-3}{z-1} \right] = \frac{\pi}{2}$. 3
Describe this locus geometrically.
- (d) If $z = \cos \theta + i \sin \theta$:
- (i) Show that $z^n + \frac{1}{z^n} = 2 \cos n\theta$ 2
- (ii) Let $\omega = z + \frac{1}{z}$. Prove that $\omega^3 + \omega^2 - 2\omega - 2 = z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3}$ 2
- (iii) Hence solve $\cos \theta + \cos 2\theta + \cos 3\theta = 0$ for $0 \leq \theta \leq 2\pi$ 2

End of Assessment Task

2010 YEAR 12 X2 TASK 1 SOLUTIONS

Question 1:

a) $z = 3 - 4i$

i) $z^3 = 3^3 + 3(3)^2(-4i) + 3(3)(-4i)^2 + (-4i)^3$
 $= 27 - 108i - 144 + 64i$
 $= -117 - 44i$ ✓

ii) $z - 2\bar{z} = (3 - 4i) - 2(3 + 4i)$
 $= -3 - 12i$ ✓

iii) $\frac{-i}{z} = \frac{-i}{3 - 4i} \times \frac{3 + 4i}{3 + 4i}$ ✓
 $= \frac{4 - 3i}{25}$
 $= \frac{4}{25} - \frac{3}{25}i$ ✓

b) let $z = x + iy$

$\therefore (x + iy)^2 = 5 + 12i$

so $x^2 - y^2 + 2xyi = 5 + 12i$ ✓

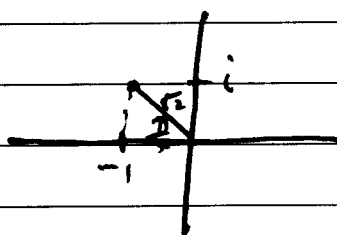
Equating real and imaginary parts.

$x^2 - y^2 = 5$ $xy = 6$ ✓

$\therefore x = \pm 3$ $y = \pm 2$

so $z = 3 + 2i$ or $-3 - 2i$ ✓

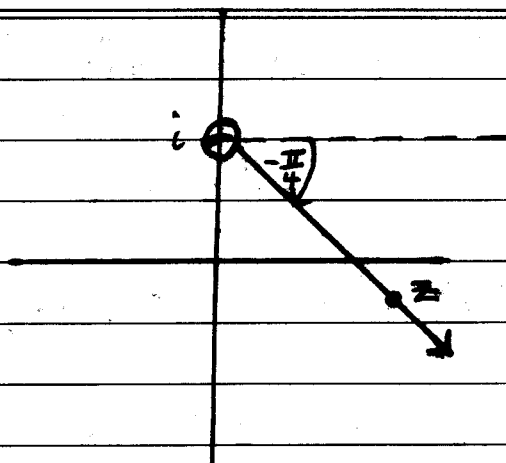
c) i)



$-1 + i = \sqrt{2} \operatorname{cis} \left(\frac{3\pi}{4} \right)$ ✓

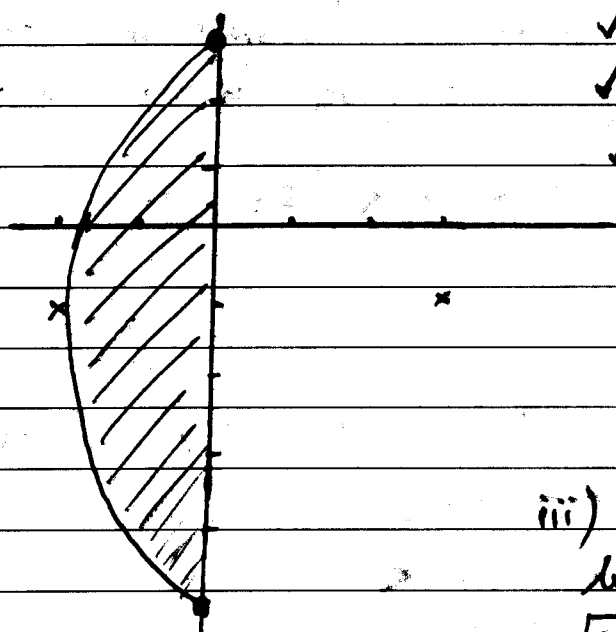
ii) $(-1 + i)^{12} = \left(\sqrt{2} \operatorname{cis} \frac{3\pi}{4} \right)^{12} = 2^6 \operatorname{cis} 9\pi = 64 \operatorname{cis} \pi$
 $= -64$ ✓

d) i)



✓ origin of ray at i not filled in
✓ direction

ii)



✓ circle centre $(3, -i)$
✓ $x \leq 0$
✓ intersection

iii) $|z| = \operatorname{Im}(z)$

let $z = x + iy$

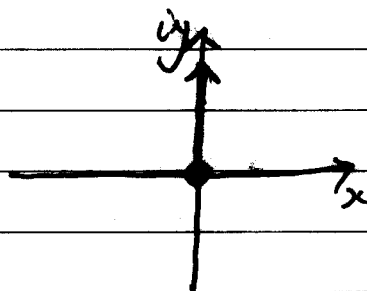
$$\sqrt{x^2 + y^2} = y$$

$$x^2 + y^2 = y^2$$

$$x = 0$$

but $|z| \geq 0$

$$\operatorname{Im}(z) = y \geq 0$$



e) To Prove $12^n > 5^n + 7^n$ $n \geq 2$

$$\text{ie } 12^n - (5^n + 7^n) > 0$$

$$\text{So when } n = 2 \quad 12^n - (5^n + 7^n) = 12^2 - (5^2 + 7^2) \\ = 70 > 0 \quad \checkmark$$

$$\therefore 12^n > 5^n + 7^n \text{ when } n = 2$$

Assume the statement is true for $n = k$
some fixed positive integer.

$$\text{ie } 12^k > 5^k + 7^k$$

When $n = k+1$

$$\begin{aligned} 12^n - (5^n + 7^n) &= 12^{k+1} - (5^{k+1} + 7^{k+1}) \\ &= 12 \cdot 12^k - 5 \cdot 5^k - 7 \cdot 7^k \\ &\geq 12(5^k + 7^k) - 5 \cdot 5^k - 7 \cdot 7^k \\ &= 12 \cdot 5^k - 5 \cdot 5^k + 12 \cdot 7^k - 7 \cdot 7^k \\ &= 7 \cdot 5^k + 5 \cdot 7^k > 0 \end{aligned}$$

as $5^k, 7^k > 0$
for all k

If the statement is true for $n = k$

Therefore the statement is true for $n = k+1$

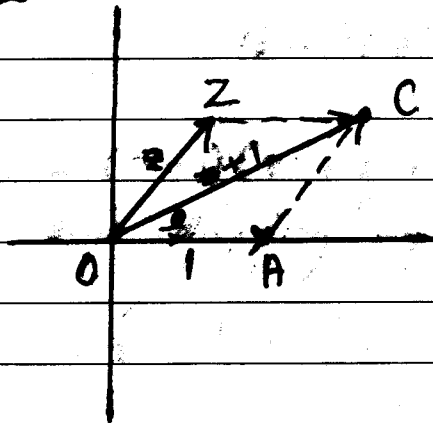
Since the statement is true for $n = 1$

Then the statement is true for $n = 2, 3, 4, \dots$

$$\therefore 12^n > 5^n + 7^n \text{ for } n \geq 2.$$

Question 2

a) i)



Since $|z| = 1$, $|OZ| = |OA| = |ZC| = |AC|$ ✓

∴ OACZ is a Rhombus (All sides equal)

ii) Since OACZ is a rhombus

$OC \perp AZ$ (diagonals bisect at 90°) ✓

∴ $\vec{AZ} = ki \vec{OC}$ (anticlockwise rotation of 90°) ✓

ie, $z - 1 = ki(z + 1)$ ✓

∴ $\frac{z-1}{z+1} = ki$ which is purely imaginary.

iii) $|z+1| = 1^2 + 1^2 - 2(1)(1)\cos(180-\theta)$ ✓
 $= 2 + 2\cos\theta$ ✓

$\arg(z+1) = \frac{\theta}{2}$ ✓

$$b) \quad i) \quad z^9 - 1 = (z^3 - 1)(z^6 + z^3 + 1) \quad \checkmark$$

$$= (z - 1)(z^2 + z + 1)(z^6 + z^3 + 1)$$

$$\text{and } z^9 - 1 = (z - 1)(z^8 + z^7 + z^6 + z^5 + z^4 + z^3 + z^2 + z + 1)$$

Equating and dividing by $z - 1 \neq 0$

$$1 + z + z^2 + \dots + z^8 = (z^2 + z + 1)(z^6 + z^3 + 1) \quad \checkmark$$

$$ii) \quad \text{For } z^9 - 1 = 0$$

$$z_k = \text{cis } \frac{2k\pi}{9} \quad k = 0, 1, 2, \dots, 8 \quad \checkmark$$

$$z^2 + z + 1 = 0$$

$$z = \frac{-1 \pm \sqrt{-3}}{2} = -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i \quad \checkmark$$

Solutions of $z^3 - 1 = 0$ are $1, \text{cis } \frac{2\pi}{3}, \text{cis } -\frac{2\pi}{3}$ $\checkmark$

So remaining solutions of solutions of

$$z^6 + z^3 + 1 = 0 \quad \text{are } \text{cis } \pm \frac{2\pi}{9}, \text{cis } \pm \frac{4\pi}{9}, \text{cis } \pm \frac{8\pi}{9} \quad \checkmark$$

$$iii) \quad \text{Sum of roots} = 0$$

$$\therefore \text{cis } \frac{2\pi}{9} + \text{cis } \frac{4\pi}{9} + \text{cis } \frac{8\pi}{9} + \text{cis } -\frac{8\pi}{9} + \text{cis } -\frac{4\pi}{9} + \text{cis } -\frac{2\pi}{9} = 0$$

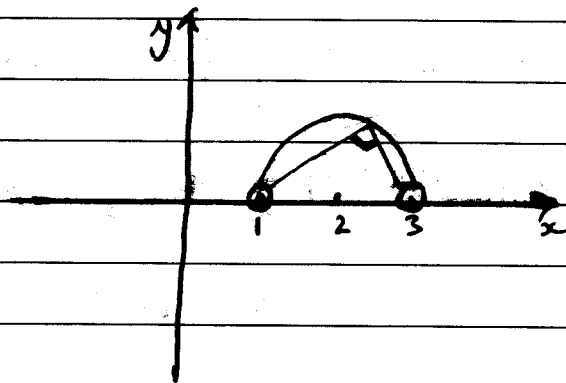
$$\text{ie, } 2 \left(\cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} + \cos \frac{8\pi}{9} \right) = 0 \quad \checkmark \quad (\text{equating real parts})$$

$$\text{so } \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = -\cos \frac{8\pi}{9}$$

$$\text{so } \cos \frac{2\pi}{9} + \cos \frac{4\pi}{9} = \cos \frac{\pi}{9} \quad \checkmark$$

$$\text{as } \cos(\pi - \theta) = -\cos \theta$$

c) Semi Circle Centre $(2, 0)$ radius 1 unit. ✓ describe



✓ sketch

✓ open circles
at $x=1$ or $x=3$

d) i) if $z = \cos \theta + i \sin \theta$

$$z^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta)$$

$$= \cos(n\theta) - i \sin(n\theta)$$

but $\cos(-\alpha) = \cos(\alpha)$

$\sin(-\alpha) = -\sin \alpha$

$$\therefore z^n + z^{-n} = 2 \cos n\theta$$

ii) L.H.S = $w^3 + w^2 - 2(w+1)$

$$= w^2(w+1) - 2(w+1)$$

$$= (w^2 - 2)(w+1)$$

$$= \left[\left(z + \frac{1}{z} \right)^2 - 2 \right] \left[z + \frac{1}{z} + 1 \right]$$

$$= \left(z^2 + \frac{1}{z^2} + 2 - 2 \right) \left(z + \frac{1}{z} + 1 \right)$$

$$= \left(z^2 + \frac{1}{z^2} \right) \left(z + \frac{1}{z} + 1 \right)$$

$$= z^3 + z + z^2 + \frac{1}{z} + \frac{1}{z^3} + \frac{1}{z^2}$$

$$= z + \frac{1}{z} + z^2 + \frac{1}{z^2} + z^3 + \frac{1}{z^3}$$

$$\text{iii)} \quad \cos \theta + \cos 2\theta + \cos 3\theta = 0 \quad 0 \leq \theta \leq 2\pi$$

$$\text{if} \quad \cos n\theta = \frac{1}{2} \left(z^n + \frac{1}{z^n} \right) \quad \checkmark$$

$$\text{then} \quad \frac{1}{2} \left(z + \frac{1}{z} \right) + \frac{1}{2} \left(z^2 + \frac{1}{z^2} \right) + \frac{1}{2} \left(z^3 + \frac{1}{z^3} \right) = 0$$

$$\text{ie,} \quad \frac{1}{2} (w^3 + w^2 - 2w - 2) = 0 \quad \text{from part ii)}$$

$$\text{so.} \quad w^2(w+1) - 2(w+1) = 0$$

$$(w^2 - 2)(w+1) = 0$$

$$w = -1, \pm\sqrt{2}$$

$$\text{but} \quad w = z + \frac{1}{z} = 2\cos\theta$$

$$\therefore \quad 2\cos\theta = -1 \quad 2\cos\theta = \pm\sqrt{2}$$

$$\cos\theta = -\frac{1}{2}$$

$$\cos\theta = \pm\frac{1}{\sqrt{2}}$$

$$\therefore \theta = \frac{\pi}{4}, \frac{2\pi}{3}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{4\pi}{3}, \frac{7\pi}{4} \quad \checkmark$$